



QATIP Intermediate

Azure Lab03a

Service Principles

Contents

Overview	1
Before You Begin	2
Phase 0 – Create a Limited-Service Principal and Set Environment Variables .	2
Purpose	2
Steps	3
Phase 1 – Run Terraform With Limited SP Identity	4
Purpose	4
Steps	4
Phase 2 – Assign Contributor Role to the Service Principal.....	4
Purpose	4
Steps	5

Overview

This lab introduces secure, least-privilege access using **Azure Service Principals (SPs)**. You will:

- Create a Service Principal with limited permissions scoped to the subscription
- Use SP credentials via environment variables

- Run Terraform operations and observe RBAC restrictions
- Upgrade permissions and reattempt deployments

Before You Begin

1. Ensure you have Owner or User Access Administrator permissions at the **subscription** level.
2. Azure CLI and Terraform must be installed.
3. Navigate to the lab directory:

```
cd c:\azure-tf-int\labs\03a
```

Phase 0 – Create a Limited-Service Principal and Set Environment Variables

Purpose

What we are doing:

We are creating a new **non-human identity** (Service Principal) that will be used to run Terraform, representing a **limited automation identity** — like what you would use in CI/CD pipelines.

Why it matters:

Rather than using our full-access lab account, we want to simulate **realistic access boundaries** — mimicking how secure organizations prevent developers and automation tools from having unrestricted rights.

Environment variable usage:

Instead of hardcoding secrets, we inject them into the environment. This aligns with DevSecOps best practices and is supported natively by the Terraform AzureRM provider.

Steps

1. Navigate to **terraform.tfvars** and substitute <random unique suffix> with random letters and/or numbers of your choice to ensure name uniqueness.
2. Update line 2 of **main.tf**, replacing <your subscription id> with your lab subscription id.
3. Create the Service Principal scoped to your subscription (replace <sub_id> with your lab subscription and <3 digits> with 3 digits of your choice):

```
az ad sp create-for-rbac --name terraform-sp-<3 digits> --role Reader  
--scopes /subscriptions/<sub_id>
```

4. Capture the following values from the output:

```
appId → ARM_CLIENT_ID  
password → ARM_CLIENT_SECRET  
tenant → ARM_TENANT_ID  
subscriptionId → ARM_SUBSCRIPTION_ID
```

5. Set environment variables:

```
$env:ARM_CLIENT_ID = "<appId>"  
$env:ARM_CLIENT_SECRET = "<password>"  
$env:ARM_SUBSCRIPTION_ID = "<sub_id>"  
$env:ARM_TENANT_ID = "<tenant_id>"
```

6. Adopt the SP identity:

```
az logout  
  
az login --service-principal --username $env:ARM_CLIENT_ID --  
password=$env:ARM_CLIENT_SECRET --tenant $env:ARM_TENANT_ID
```

Then run:

```
az account show
```

You should see "type": "servicePrincipal" in the output.

Phase 1 – Run Terraform With Limited SP Identity

Purpose

What we are doing:

Attempting to deploy infrastructure using Terraform under this limited SP identity.

Why it matters:

This phase is expected to fail — and that is intentional. It highlights that simply authenticating to Azure is not enough; **role assignments define what you are allowed to do**. It also encourages developers to review error messages meaningfully and connect failures to permissions.

Steps

1. Run:

```
terraform init
terraform plan
terraform apply
```

2. Expect Terraform apply to **fail** with authorization errors.

Phase 2 – Assign Contributor Role to the Service Principal

Purpose

What we are doing:

Temporarily switching back to the lab account to assign Contributor rights to the SP.

Why it matters:

This demonstrates the **RBAC model in action** — where a principal's capabilities change dynamically based on role assignments. It mirrors how platform teams might approve temporary elevated access to unblock automation or development pipelines.

Steps

1. Re-adopt your lab identity:

```
az logout
```

```
az login
```

Enter your student credentials in the launched browser session

2. Inserting your SP id and Subscription id, run:

```
az role assignment create --assignee <appid> --role Contributor  
--scope /subscriptions/<sub_id>
```

3. Re-adopt the SP identity:

```
az logout
```

```
az login --service-principal --username $env:ARM_CLIENT_ID --  
password=$env:ARM_CLIENT_SECRET --tenant $env:ARM_TENANT_ID
```

4. Retry:

```
terraform plan
```

```
terraform apply
```

5. Terraform should now **succeed**.
6. Run **terraform destroy** to remove the resources.
7. Delete the Service Principal : **az ad sp delete --id <appid>**

Congratulations, you have completed this lab

Copyright

© 2026 QA Michael Coulling-Green. All rights reserved. These lab materials are provided as part of an authorised training course. Course attendees may reuse these materials for their own personal learning and reference outside of the training session. Unauthorised copying, redistribution, publication, or use of these materials to deliver training is prohibited without prior written permission from the author.

© QA 2026